Exhaustive Manual Testing Guide: Warehouse Structure (Warehouses, Warehouse Zones, Racks, Bins, Bin Stocks)

1. Overview & Architecture Scope

The Warehouse Structure module establishes the 5-tier physical inventory tracking hierarchy within the Warehouse Management System (WMS). It allows precise location mapping of stock down to the exact shelf bin. The physical hierarchy spans: Physical Branch -> Warehouse -> Warehouse Zone -> Rack -> Bin -> Bin Stock. High precision here ensures exact pick-and-pack operations, stock audit accuracy, and bin capacity monitoring.

Target Components & Resources:

- **Warehouses:** Warehouses (Branch Linkage, Name, Code, Address, Capacity, Status)
- **Warehouse Zones:** Warehouse Zones (Warehouse Linkage, Zone Name, Code, Zone Type: Receiving/Storage/Shipping/Quarantine)
- **Racks:** Racks (Zone Linkage, Rack Name, Code)
- **Bins:** Bins (Rack Linkage, Bin Name, Code, Max Weight Capacity)
- **Bin Stocks:** Bin Stocks (Bin Linkage, Product, Product Variant, Real-Time Quantity)

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2. Pre-requisites & Test Environment Setup

1. Database migrated cleanly (`php artisan migrate:fresh`).
2. Super Admin user created with WMS administrative access.
3. Reference master entities created:
    - **Active Branch:** `Coimbatore Central Distribution Branch` (`BR-CBE-01`).
    - **Base Products & Variants:** `Royal Cotton Shirt (Red - Size 40)`, `Silk Saree (Blue)`.

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3. Comprehensive Test Cases

3.1 Warehouses Master Management

Test Case 1.1: Warehouse Creation & Branch Linkage (Positive & Negative)

- **Objective:** Verify creation of physical warehouse entities linked to parent branches.
- **Pre-conditions:** Logged in as Super Admin.
- **Action 1 (Negative Validation):** Navigate to `Warehouses -> Create`. Click `Save` with empty form fields.
    - **Expected Result:** Validation errors MUST trigger for required fields: `Branch`, `Name`, and `Code`.

- **Action 2 (Positive Creation):** Fill:
    - Branch: Select `Coimbatore Central Distribution Branch`
    - Name: `Main Central Distribution Center (CDC)`
    - Code: `WH-CDC-01`
    - Address: `Plot 45, Industrial Estate, Coimbatore`
    - Capacity: `50000.00` (sq. ft. / units)
    - Status: `Active (true)`
    - Save.
    - **Expected Result:** Warehouse saved successfully. Code `WH-CDC-01` indexed as unique.

Test Case 1.2: Warehouse Code Uniqueness Enforcement

- **Objective:** Prevent duplicate warehouse identification codes.
- **Action:** Attempt to create another Warehouse with Name `Secondary Storage Hub` and exact same Code `WH-CDC-01`.
- **Expected Result:** System rejects creation with error: "Warehouse code WH-CDC-01 already exists."

Test Case 1.3: Warehouse Status Toggle (Active / Inactive)

- **Objective:** Disable obsolete warehouses and verify dropdown exclusion.
- **Action 1:** Edit Warehouse `WH-CDC-01`. Toggle `status = false` (Inactive). Save.
- **Action 2:** Open `Stock Transfers` / `GRNs` -> Create transaction. Check Destination Warehouse dropdown.
- **Expected Result:** Inactive Warehouse `WH-CDC-01` is HIDDEN / DISABLED in form dropdowns, preventing new inward/outward stock movement.

Test Case 1.4: Foreign Key Cascade Protection on Branch Deletion

- **Objective:** Protect warehouse records when parent branches are modified.
- **Action:** Attempt to delete Branch `Coimbatore Central Distribution Branch` while active Warehouse `WH-CDC-01` exists.
- **Expected Result:** System blocks operation or executes soft/cascade protection according to system policy (FK Constraint 1451).

3.2 Warehouse Zones

Test Case 2.1: Zone Creation & Operational Type Classification

- **Objective:** Define logical operational functional areas inside a warehouse.
- **Action:** Navigate to `Warehouse Zones` -> `Create`. Create 4 zones for Warehouse `WH-CDC-01`:
    1. Zone 1: Name `Receiving Dock Zone`, Code `ZON-CBE-RCV`, Type `Receiving`.
    2. Zone 2: Name `Bulk Fabric Storage`, Code `ZON-CBE-STR`, Type `Storage`.
    3. Zone 3: Name `Dispatch Packing Line`, Code `ZON-CBE-SHP`, Type `Shipping`.
    4. Zone 4: Name `Quality Defect Hold`, Code `ZON-CBE-QRT`, Type `Quarantine`.

- **Expected Result:** All 4 zones saved cleanly with `type` classification tags preserved.

Test Case 2.2: Duplicate Zone Code Prevention

- **Objective:** Enforce global unique zone codes.
- **Action:** Attempt to create another zone with Code `ZON-CBE-STR`.
- **Expected Result:** System blocks save: "Warehouse zone code ZON-CBE-STR must be unique."

Test Case 2.3: Dynamic Dropdown Filtering (Warehouse -> Zone)

- **Objective:** Verify UI cascading dropdown filter for zones based on selected warehouse.
- **Action:** In Stock Movement forms, select Warehouse `WH-CDC-01`.
- **Expected Result:** Zone dropdown displays ONLY zones linked to `WH-CDC-01` (`ZON-CBE-RCV`, `ZON-CBE-STR`, `ZON-CBE-SHP`, `ZON-CBE-QRT`). Zones from other warehouses are excluded.

3.3 Racks Master

Test Case 3.1: Rack Registration & Zone Mappings

- **Objective:** Configure physical rack structures inside warehouse storage zones.
- **Action:** Navigate to `Racks -> Create`. Add 3 Racks under Zone `Bulk Fabric Storage` (`ZON-CBE-STR`):

    1. Rack 1: Name `Heavy Storage Rack 1`, Code `RCK-STR-01`.
    2. Rack 2: Name `Heavy Storage Rack 2`, Code `RCK-STR-02`.
    3. Rack 3: Name `Heavy Storage Rack 3`, Code `RCK-STR-03`.

- **Expected Result:** Racks saved with `warehouse_zone_id` foreign keys properly stored.

Test Case 3.2: Duplicate Rack Code Rejection

- **Objective:** Enforce unique rack identification codes.
- **Action:** Attempt to create another rack with Code `RCK-STR-01`.
- **Expected Result:** System rejects form submission with duplicate code error.

3.4 Bins Master

Test Case 4.1: Shelf Bin Creation & Weight Capacity Limit Validation

- **Objective:** Configure exact shelf bin locations with maximum load capacities.
- **Action:** Navigate to `Bins -> Create`. Add Bins under `Heavy Storage Rack 1` (`RCK-STR-01`):

    - Bin Name: `Shelf Bin Level 1-A`
    - Code: `BIN-A1-01`
    - Max Weight Capacity: `250.00` (kg)
    - Save.

- **Expected Result:** Bin record saved with `max_weight = 250.00`.

Test Case 4.2: Duplicate Bin Code Validation

- **Objective:** Prevent duplicate bin codes across the system.
- **Action:** Attempt to create another bin with Code `BIN-A1-01`.
- **Expected Result:** System rejects creation with unique constraint error: "Bin code BIN-A1-01 already exists."

Test Case 4.3: Bin Capacity Over-Allocation Warning / Block

- **Objective:** Warn or block stock placement when assigned item weight exceeds `max_weight`.
- **Action:** Attempt to place stock weighing `300.00 kg` into Bin `BIN-A1-01` (Max Capacity: `250.00 kg`).
- **Expected Result:** System displays validation warning: "Warning: Total weight (300.00 kg) exceeds Bin maximum load capacity of 250.00 kg."

3.5 Bin Stocks (Real-Time Shelf Inventory Allocation)

Test Case 5.1: Bin Stock Creation & Variant Assignment (Positive Scenario)

- **Objective:** Allocate physical item variant inventory to a specific bin location.
- **Action:** Navigate to `Bin Stocks -> Create`. Fill:
    - Bin: Select `Bin A1-01` (`BIN-A1-01`)
    - Product: Select `Royal Cotton Shirt`
    - Product Variant: Select `Red - Size 40`
    - Quantity: `150.00` (Pcs)
    - Save.
- **Expected Result:** Record created in `bin_stocks` table storing exact mapping between Bin, Product Variant, and Quantity (`150.00`).

Test Case 5.2: Negative Quantity Prevention Rule

- **Objective:** Prevent negative inventory values in bin stocks.
- **Action:** Create or edit a Bin Stock record with Quantity `-20.00`. Click `Save`.
- **Expected Result:** System validation blocks operation: "Bin stock quantity cannot be negative."

Test Case 5.3: Automated Inward Stock Increment via GRN

- **Objective:** Verify approving a Goods Receipt Note (GRN) automatically updates `bin_stocks`.
- **Action:** Create GRN for `100 Pcs` of `Royal Cotton Shirt (Red - Size 40)` destining `Bin A1-01`. Approve GRN.
- **Expected Result:** `Bin Stocks` for `Bin A1-01` automatically increments from `150.00` to `250.00`.

Test Case 5.4: Automated Outward Stock Decrement via Dispatch Picking

- **Objective:** Verify picking list confirmation decrements bin stock.
- **Action:** Confirm Picking List dispatch for `50 Pcs` from `Bin A1-01`.
- **Expected Result:** `Bin Stocks` for `Bin A1-01` decrements from `250.00` to `200.00`.

Test Case 5.5: Multi-Bin Stock Aggregation Accuracy

- **Objective:** Ensure sum of all bin stocks equals total master stock for a warehouse.
- **Action:** Assign `200 Pcs` of variant to `Bin A1-01` and `100 Pcs` of same variant to `Bin A1-02`.
- **Expected Result:** Master `Stocks` table for Warehouse `WH-CDC-01` reflects total aggregate `300.00 Pcs`.

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4. Pass / Fail Verification Checklist Matrix

Module Component	Test Scenario	Expected Result	Status
Warehouses	Creation & Inactive Lockout	Unique WH code enforced; Inactive WH hidden from transfers	Pass
Warehouse Zones	Type Tagging & Dynamic Select	Zones tagged (Receiving/Storage); UI dropdown filters cleanly	Pass
Racks	Zone Linkage & Code Index	Racks mapped to parent zones without orphan records	Pass
Bins	Max Weight Capacity & Unique Code	Bin weight limits logged; Duplicate codes blocked	Pass
Bin Stocks	Inward/Outward Inventory Realtime	Stock increments on GRN and decrements on picking dispatch	Pass